

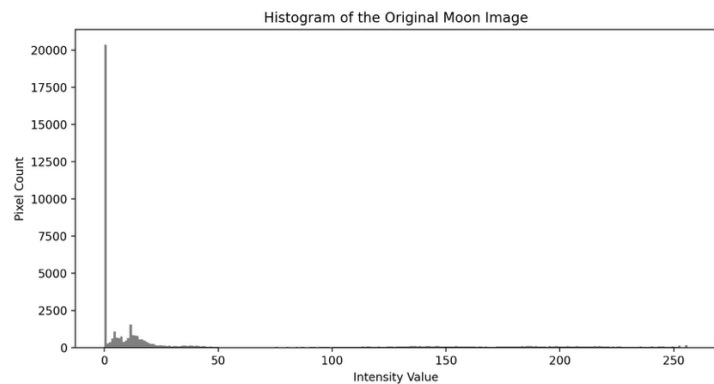
Homework 1: Histogram Equalization and Specification

1. Procedures for Preparing Test Images

To evaluate the effectiveness of image enhancement algorithms, four types of degraded images were generated from the original grayscale "moon" image:



Fig. 1. Original Image



- **Dark Image:** Generated by scaling down all pixel intensities by a factor of 0.3. This shifts the histogram to the far left (low intensity range).
- **Bright Image:** Created by adding a constant offset of 100 to each pixel (clamped at 255). The histogram is shifted toward the right.
- **Low Contrast Image:** Produced by compressing the dynamic range into a narrow interval (100 to 150). This results in a "washed out" appearance with a spiked histogram.
- **High Contrast Image:** Generated using linear stretching on the range [50, 200], pushing intensities toward the extremes (0 and 255).

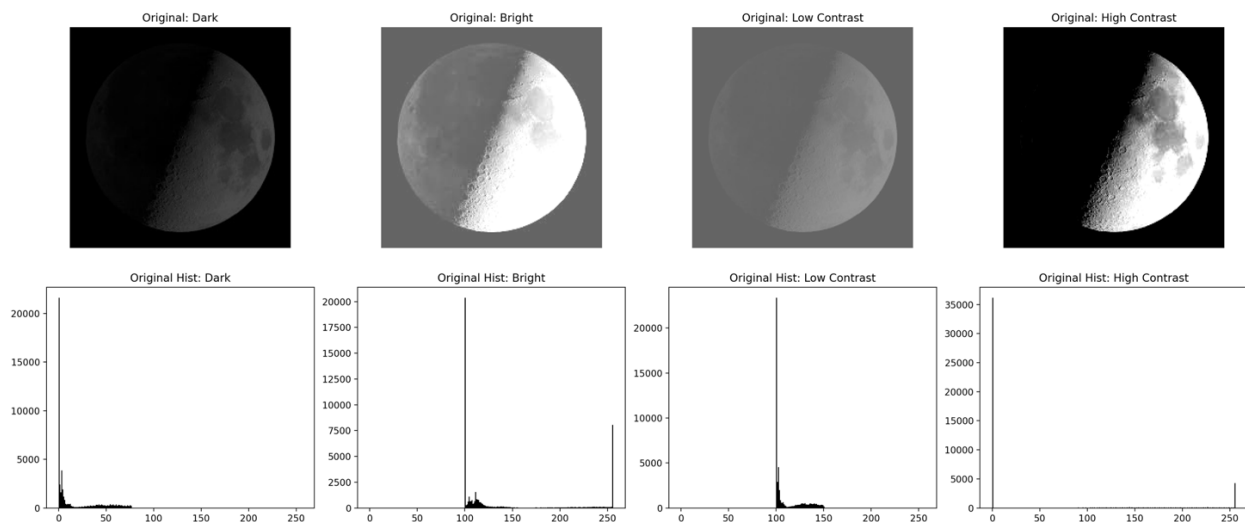


Fig. 2. Test Images and Corresponding Histogram

2. Description of Methods

The primary method used for enhancement is **Histogram Equalization (HE)**, implemented manually without built-in library functions:

A. Histogram Equalization (HE)

The goal of HE is to find a transformation $T(r)$ such that the output histogram is approximately uniform. The transformation is based on the **Cumulative Distribution Function (CDF)**:

$$s_k = (L - 1) \sum_{j=0}^k p_r(r_j)$$

where:

- L is the total number of gray levels (256).
- $p_r(r_j)$ is the probability of occurrence of gray level j .

B. Power Law (Gamma) Transformation

$$s = c \cdot r^\gamma$$

Applying $\gamma < 1$ (e.g., 0.4) helps in enhancing details in dark regions more smoothly than HE.

3. Experimental Results and Discussion

Analysis of Results:

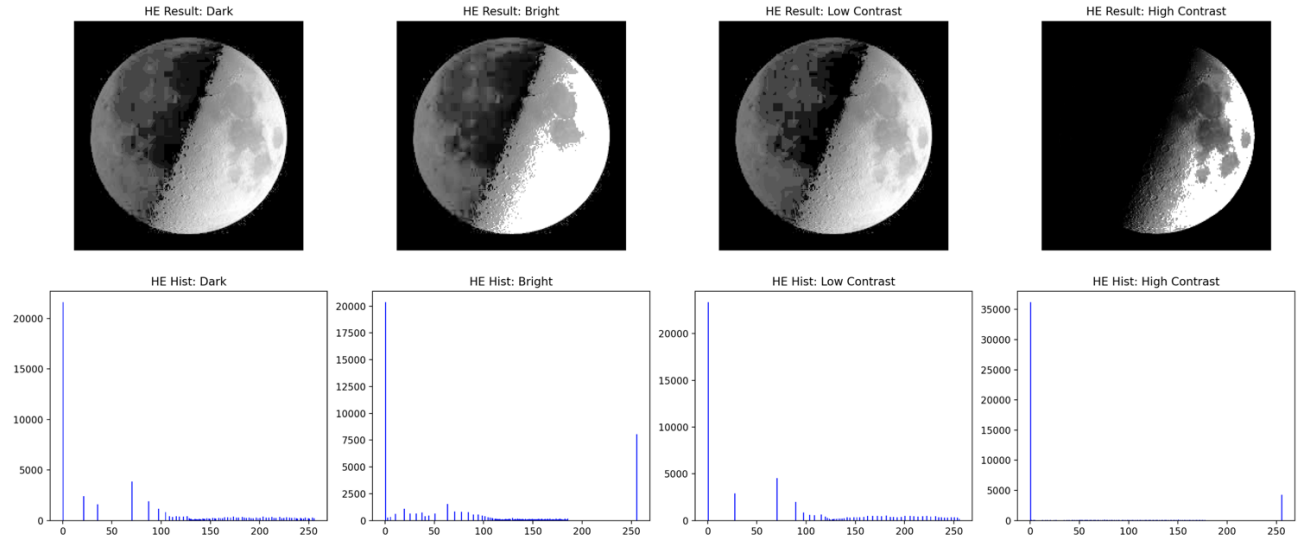


Fig. 3. Histogram Equalization Result

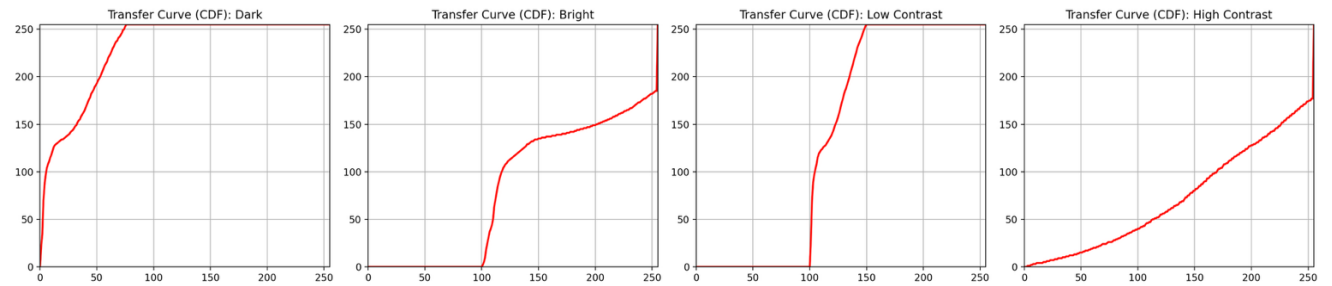


Fig. 4. Transfer Curve

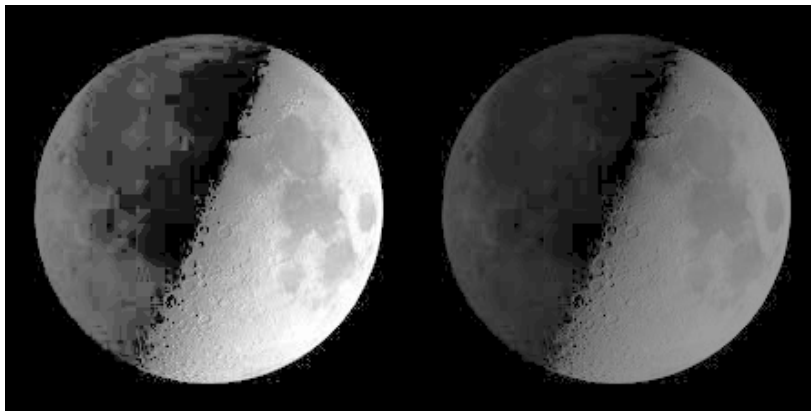


Fig. 5. HE vs Power Law

The experimental results demonstrate that **Histogram Equalization (Fig. 3)** is highly effective for images with narrow intensity distributions. As shown in the **Low Contrast** case, HE successfully spreads the pixel values across the entire range $[0, 255]$,

significantly revealing surface details of the moon that were previously invisible. The **Transfer Curve (Fig. 4)** for the Low Contrast image exhibits a steep slope in the middle intensity range, which directly corresponds to this dramatic expansion of contrast.

However, HE also reveals significant limitations when applied to **Dark** or **Bright** images. In the Dark image enhancement, noticeable **contouring artifacts** and amplified background noise are observed. This occurs because the algorithm maps a very narrow set of input gray levels to a wide output range, causing "gaps" in the resulting histogram and destroying smooth gradients. In contrast, the **Power Law (Fig. 5) Transformation** ($\gamma < 1$) provides a much more natural and smoother enhancement for dark regions. Unlike the automatic nature of HE, Power Law allows for manual control over the intensity mapping, preventing the over-amplification of noise and preserving the visual integrity of the image.

4. Conclusion

In conclusion, **Histogram Equalization** is a robust and automatic tool for enhancing images with concentrated histograms, but it is not a "one-size-fits-all" solution. It excels at improving global contrast but often leads to over-enhancement and artifacts in images with extreme brightness levels or low initial bit-depth. For tasks requiring more nuanced control, such as preserving specific tonal qualities or preventing noise in dark backgrounds, methods like **Power Law transformations** or **Histogram Specification** (which will be explored in the next section) offer superior results by allowing the user to define or influence the target distribution.